UP Special Topic 7 — Transport, Tourism Circuits, Forests, Protected Areas, Wetlands and Disaster Management

Current Affairs

Year	Fact	Why it matters
Rolling	Refresh edition-bound figures from latest Census / Survey / State CA desks.	Do not invent interim totals.

Teaching cards

2.1 Forest Cover, Forestry Statistics & ISFR Data for Uttar Pradesh

A. Forest and Tree Cover Profile (India State of Forest Report - ISFR)

- **Total Recorded Forest Area (RFA)**: 17,384 km² (approx. 7.22% of UP's geographical area).
- **Actual Forest Cover (ISFR Data)**:
- Total Forest Cover = **14,818 km²** (**6.15% of the total geographical area of UP**).
- *Breakdown*:
  - Very Dense Forest (VDF): **1,096 km²** (0.45%)
  - Moderately Dense Forest (MDF): **4,009 km²** (1.66%)
  - Open Forest (OF): **9,713 km²** (4.03%)
- **Tree Cover (Trees outside forest)**: **7,421 km²** (**3.08% of total area**).
- **Total Green Cover (Forest + Tree Cover)**: **22,239 km²** (**9.23% of total area**).
- *National Target*: National Forest Policy (1988) mandates a 33.3% forest/tree cover target; UP currently operates at ~9.23%, driving aggressive afforestation campaigns (*Vriksharopan Jan Andolan*).

B. District-Wise Forest Cover Extremes

FOREST COVER EXTREMES IN UTTAR PRADESH		
Parameter	Top Districts (Highest)	Bottom Districts (Lowest)
1. By Absolute Forest Area	1. **Sonbhadra** (2,528 km²) 2. Lakhimpur Kheri (1,273 km²) 3. Mirzapur (758 km²) 4. Pilibhit (687 km²)	1. **Bhadohi** (3.71 km²) 2. Mau (11.00 km²) 3. Mainpuri (13.64 km²) 4. Sant Kabir Nagar (14.0 km²)
2. By Percentage of District	1. **Sonbhadra (35.29%)** 2. **Chandauli (21.78%)**	1. **Bhadohi (0.37%)** 2. Mainpuri (0.49%)

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|    Area          | 3. Chitrakoot (19.64%)    | 3. Deoria (0.60%)              |
|                  | 4. Mirzapur (17.20%)      | 4. Badaun (0.62%)              |
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> **Trap**: Notice that only **Sonbhadra (35.29%)** and **Chandauli (21.78%)** have more than **20% forest cover** in Uttar Pradesh. Districts like **Bhadohi, Mainpuri, and Deoria** have less than 1% forest cover.

2.2 Wildlife Protection, Tiger Reserves and Conservation Infrastructure

A. The Four Tiger Reserves of Uttar Pradesh

1. **Dudhwa Tiger Reserve (Lakhimpur Kheri & Bahraich)**: - Notified under Project Tiger in **1987–88**. - Includes Dudhwa National Park, Kishanpur Sanctuary, and Katarniaghat Sanctuary. - Rich in swamp deer (over 1,500 barasingha), tigers, Asian elephants, and reintroduced Indian one-horned rhinoceros.
2. **Pilibhit Tiger Reserve (Pilibhit)**: - Notified in **2014**; sits at the base of Siwalik foothills. - Drained by Sharda, Gomti, and Khakra rivers. Won the global **TX2 Award in 2020** for doubling wild tiger numbers from 25 to 65 within a decade.
3. **Amangarh Tiger Reserve (Bijnor)**: - Notified in **2012**; covers 95 km² on the border of Uttarakhand. - Functions as an essential buffer corridor for Corbett National Park tigers and elephants.
4. **Ranipur Tiger Reserve (Chitrakoot)**: - Notified in **October 2022** as India's 53rd Tiger Reserve and UP's 4th Tiger Reserve. - Covers 529.8 km² in the Bundelkhand Vindhyan hill tract, serving as a biological bridge for tigers migrating from Madhya Pradesh's Panna Tiger Reserve.

B. Elephant Conservation (Project Elephant)

- **Uttar Pradesh Elephant Reserve (UPER)**: Notified in **2009**, spanning 744 km² across Saharanpur and Bijnor (Shivalik Elephant Reserve).
- **Terai Elephant Reserve (TER)**: Notified in **2022** as the 33rd Elephant Reserve in India, covering 3,049 km² across Dudhwa, Katarniaghat, and Kishanpur tracts.

2.3 Tourism Circuits & Cultural Heritage Economy

The Uttar Pradesh Tourism Policy structures state tourism promotion into specialized thematic circuits:

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|                   THEMATIC TOURISM CIRCUITS IN UP                    |
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| Circuit Name     | Principal Destinations & Historical Heritage      |
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| 1. Buddhist      | Sarnath (Varanasi), Kushinagar, Shravasti, Kaushambi,  |
|    Circuit       | Sankisa (Farrukhabad), Kapilavastu (Piprahwa).    |
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| 2. Ramayana      | Ayodhya, Nandigram, Shringverpur (Prayagraj),     |
|    Circuit       | Chitrakoot Dham.                                  |
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| 3. Krishna       | Mathura, Vrindavan, Gokul, Barsana, Nandgaon, Govardhan. |
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Circuit	
4. Heritage Arc	Agra (Taj Mahal, Fatehpur Sikri) – Lucknow (Nawabi monuments) – Varanasi (Ghats, Kashi Vishwanath Corridor).
5. Bundelkhand Circuit	Jhansi (Rani Mahal, Fort), Deogarh (Dashavatara Temple), Kalinjar Fort (Banda), Chitrakoot, Mahoba.
6. Eco-Tourism Circuit	Dudhwa NP, Pilibhit TR (Chuka Beach), Katarniaghat WLS, National Chambal Sanctuary, Chandraprabha WLS.

2.4 Environmental Pollution Hotspots and Disaster Vulnerability

A. Environmental Pollution Hotspots

1. **River Pollution (Ganga, Yamuna, Gomti)**: - *Kanpur Stretch of Ganga*: Heavy chromium and chemical effluent pollution from tanneries (Jajmau industrial cluster) and untreated municipal sewage. - *Yamuna Stretch (Delhi to Agra/Etawah)*: High biological oxygen demand (BOD) and toxic foam due to industrial discharges and lack of environmental base flow. - *Gomti River at Lucknow*: Critical urban pollution load; rejuvenated through the Gomti Riverfront interceptor drain project and sewage treatment plants (STPs).
2. **Taj Trapezium Zone (TTZ)**: - Defined geographical trapezoidal area of **10,400 km²** around the Taj Mahal covering Agra, Firozabad, Mathura, Hathras, and Etah in UP (and Bharatpur in Rajasthan). - Set up following the landmark Supreme Court ruling (*M.C. Mehta vs Union of India, 1996*) to protect the white marble of the Taj Mahal from acid rain (SO_2 and NO_x emissions from Mathura Refinery, glass foundries, and brick kilns).
3. **Air Pollution in NCR Belt**: - Ghaziabad, Noida, Greater Noida, and Meerut frequently register severe Air Quality Index (AQI) levels during post-monsoon winter inversion months due to vehicular emissions, construction dust, and stubble burning.

B. Disaster Vulnerability Matrix of UP

1. **Floods (Eastern UP & Terai)**: - Drained by snow-fed Himalayan rivers: **Ghaghara, Rapti, Gandak, Sharda, Rohini**. - Districts chronically affected: Gorakhpur, Maharajganj, Deoria, Siddharthnagar, Basti, Gonda, Balrampur, Bahraich, Shravasti, Ballia.
2. **Droughts (Bundelkhand & Southern Plateau)**: - Highly erratic monsoon, poor soil water-retention capacity of crystalline granitic/lateritic soils, and deep water tables. - Chronically drought-affected districts: **Jhansi, Jalaun, Lalitpur, Hamirpur, Mahoba, Banda, Chitrakoot**.
3. **Heat Waves (*Loo*) & Lightning**: - Severe summer heatwaves (>45^\circ ext{C} - 48^\circ ext{C}) in Bundelkhand, Agra, Mathura, and Prayagraj. - Increasing incidence of thunderstorm lightning strikes during pre-monsoon and monsoon months in Mirzapur, Sonbhadra, and Bundelkhand.
4. **Institutional Setup**: - **Uttar Pradesh State Disaster Management Authority (UPSDMA)**: Constituted under the Disaster Management Act, 2005; headed by the **Chief Minister**. Implements the *State Disaster Management Plan (SDMP)* and the early warning emergency alert system (*Sachet*).

Bilingual Terminology -- UP Transport, Tourism, Environment

English	Hindi	Key Anchor
Expressway	एक्सप्रेसवे	High-speed limited-access highway
Purvanchal Expressway	पूर्वांचल एक्सप्रेसवे	341 km; Lucknow to Ghazipur (longest expressway in UP)
Agra-Lucknow Expressway	आगरा-लखनऊ एक्सप्रेसवे	302 km 6-lane (world's widest access-controlled expressway in 2016)
Yamuna Expressway	यमुना एक्सप्रेसवे	165 km; Delhi-Agra
Bundelkhand Expressway	बुंदेलखंड एक्सप्रेसवे	296 km; Chitrakoot to Etawah
Ganga Expressway	गंगा एक्सप्रेसवे	594 km; Meerut to Prayagraj (longest under construction)
National Highway	राष्ट्रीय राजमार्ग	Centre-maintained highways; NH passes through UP
Lucknow Metro	लखनऊ मेट्रो	Operational since 2017; first metro in UP
Pradhan Mantri Gram Sadak Yojana	प्रधानमंत्री ग्राम सड़क योजना	Rural road connectivity to unconnected habitations
Wildlife Sanctuary	वन्यजीव अभयारण्य	Protected area for wildlife
Tiger Reserve	बाघ संरक्षित क्षेत्र	Under Project Tiger (NTCA)
National Park	राष्ट्रीय उद्यान	Highest protected area; no human habitation
Biodiversity Hotspot	जैव विविधता हॉटस्पॉट	Area of exceptional biodiversity + high threat
Pollution Control Board	प्रदूषण नियंत्रण बोर्ड	UPPCB = Uttar Pradesh Pollution Control Board
SDMA	राज्य आपदा प्रबंधन प्राधिकरण	State Disaster Management Authority

Extended Theory -- UP Transport Network

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UTTAR PRADESH -- EXPRESSWAY NETWORK
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Yamuna Expressway:   165 km (Delhi -- Agra)     [2012]
Agra-Lucknow:        302 km (Agra -- Lucknow)   [2016]
Purvanchal:          341 km (Lucknow -- Ghazipur) [2021]
Bundelkhand:         296 km (Chitrakoot -- Etawah) [2022]
Ganga Expressway:    594 km (Meerut -- Prayagraj) [Under construction]

RECORD: Agra-Lucknow Expressway was the world's widest
access-controlled expressway at inauguration (2016).
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GANGA EXPRESSWAY = longest expressway in India
(when complete at 594 km)
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UP's Wildlife Sanctuaries and National Parks

Protected Area	District	Specialty
Dudhwa National Park	Lakhimpur Kheri	Barasingha (swamp deer), tigers, rhinos
Katarniaghat Wildlife Sanctuary	Bahraich	Gharial, Gangetic dolphin, tigers
Pilibhit Tiger Reserve	Pilibhit	Tigers in terai; adjacent to Nepal
Chandraprabha Wildlife Sanctuary	Chandauli	Blackbuck, hyena
Sandi Bird Sanctuary	Hardoi	Migratory birds
Suhelwa Wildlife Sanctuary	Shravasti, Balrampur	Tigers; extended buffer of Dudhwa
National Chambal Sanctuary	Agra, Etawah	Gharial, Gangetic dolphin, red-crowned roofed turtle

Disaster Vulnerability of UP

Disaster Type	High-Risk Districts	Key Reason
Floods	Bahraich, Lakhimpur, Balrampur, Gonda, Gorakhpur	Ghaghra, Rapti, Rohini rivers; Nepal releases
Drought	Bundelkhand (Jhansi, Lalitpur, Banda)	Low rainfall; rocky terrain; poor water retention
Earthquakes	Uttarakhand border districts	Seismic Zone IV proximity
Lightning	All districts (UP is India's most lightning-prone State)	Flat terrain; pre-monsoon storms
Cold Wave	Purvanchal, Rohilkhand	Dense fog; North Indian cold wave

One-Liner Revision -- UP Transport, Tourism, Environment

#	Fast Fact
1	Purvanchal Expressway = **341 km** (Lucknow-Ghazipur); inaugurated Nov 2021
2	Agra-Lucknow Expressway = **302 km**; inaugurated 2016
3	Yamuna Expressway = **165 km** (Delhi-Agra); 2012

#	Fast Fact
4	Ganga Expressway = **594 km** (Meerut-Prayagraj); longest when complete
5	Bundelkhand Expressway = **296 km** (Chitrakoot-Etawah)
6	Lucknow Metro = first metro in UP (operational since **2017**)
7	Dudhwa National Park = famous for **Barasingha** and tigers
8	Pilibhit Tiger Reserve = terai forests; adjacent Nepal border
9	National Chambal Sanctuary = **Gharial** conservation
10	Katarniaghat WS (Bahraich) = Gangetic dolphin + gharial
11	UP = India's most lightning-prone State
12	Flood-prone districts = Bahraich, Lakhimpur Kheri, Gorakhpur
13	Drought-prone = **Bundelkhand** (Jhansi, Lalitpur, Banda)
14	UP State animal = **Barasingha** (swamp deer)
15	UP State bird = **Sarus Crane** (world's tallest flying bird)